A one-parameter crack detector, measured on SEM micrographs

DRAFT — not submitted. Every number is generated from measurement files by paper/make_paper.py; nothing in this document is typed by hand twice.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on the only public crack data we could find in its own modality: scanning-electron micrographs of metals and coatings with pixel-exact masks. On whole fields of Inconel 718 it reaches mean per-image F1 0.3444 and IoU 0.2137, which is 9.8 times the F1 and 11.8 times the IoU of marking every pixel. Its one constant, fitted years earlier on SEM steel, is the best of seven values tested here without being refitted.

Three results cut against the method and we report them with the first. Against seven classical detectors tuned identically on the same images it ranks third of seven — a local threshold and a ridge filter score as well or better. On an SEM coating at 0.26% crack prevalence it fails at 0.0102 precision. And cropping the same images around their cracks, as the dataset's own released test split does, raises F1 from 0.3444 to 0.4958 without changing the detector at all.

Status of this draft. Section 7 lists what is missing before submission.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales, takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. It has one free constant, mad_k. There is no training step and no model file.

Reporting such a thing honestly turns out to be harder than building it.

2 The detector

Figure 1 is the whole path. Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric with no free parameters: paired median IoU +0.2296 over thirteen specimens, p = 0.0024. Three specimens that had scored recall 0.0000 at every candidate setting ever tried were never invisible to the proposal stage; the classifier was rejecting them.

Tiles cannot stand in for frames. Rules validated on 512 px crops deleted 40–43% of

hand-marked crack when run on whole frames. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is extrapolating on the object it most needs to keep.

3 On public SEM data with pixel-exact ground truth

Two sets, the crack classes of a public cross-material microstructure collection (Zenodo 23045328, CC BY 4.0). They are the only public crack segmentation data we found in this modality; Section 7 records what else was searched and rejected.

set	material	n	crack prevalence	masks
Inconel 718	nickel superalloy	31	1.80%	anti-aliased binary
EBC	environmental barrier coating	24	0.26%	binary

Everything is scored at 0 px tolerance with mad_k chosen leave-one-image-out.

Two properties of the data had to be established before any score could be trusted. The mask encodings differ between the two sets — the coating's are strictly 0 or 255, the alloy's span the full byte range as anti-aliased binary, 94.7% exactly 0 and 4.3% at 246 or above — so the scorer checks bimodality and refuses a mask that fails it. And the images are 16-bit TIFFs: PIL's ordinary 8-bit conversion turns them nearly white, mean 254.8 against the data's 15510 of 65535, which drove the detector's recall to 0.0012 before it was caught. Taking the high byte reproduces the dataset authors' own 8-bit PNGs exactly, so that is the conversion used.

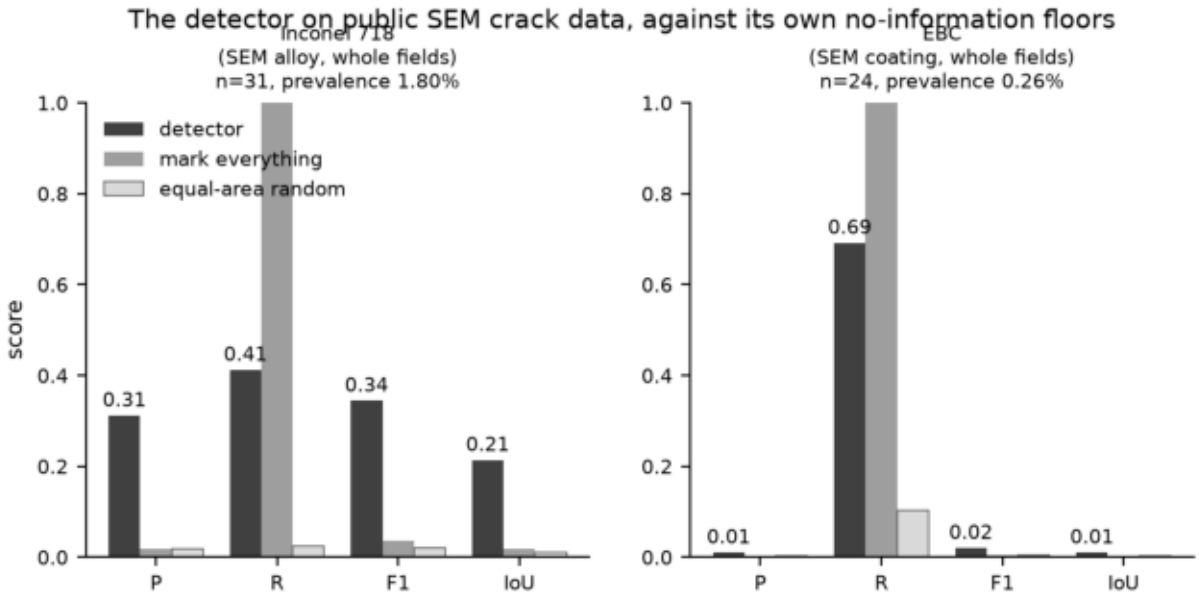


Figure 1. P1 - public scores

3.1 Whole fields, not the released crops

The collection ships a test split of 1024 px crops. We do not use it as the primary measurement for two reasons. Every one of the 18 source images behind those crops also appears among the originals, so the two are not independent sets. And the crops are cut around their cracks: prevalence rises from 1.80% to 4.53% and F1 with it, from 0.3444 to 0.4958, with no change to the detector. Figure 2 shows both curves. The difference between them is what a benchmark's framing is worth, and it is larger than most differences between methods in this literature.

3.2 The one free constant

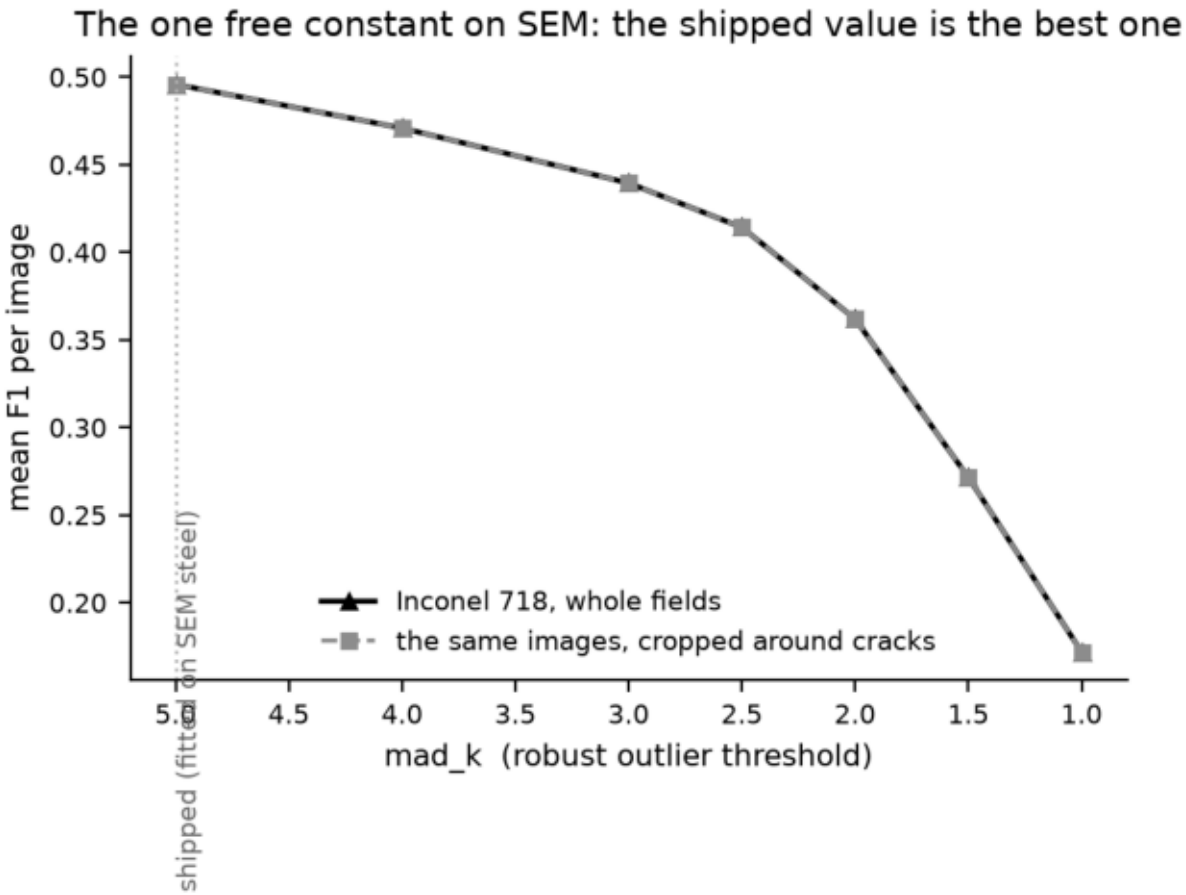


Figure 2. P2 - operating curve

mad_k = 5.0 was fitted on SEM steel in a different laboratory, on a different alloy, years before this dataset existed. On these whole fields it is the best of the seven values tested and the curve falls away monotonically below it. That is the strongest positive result here: the constant did not need refitting for a new SEM corpus.

3.3 The failure case

On the EBC coating the detector reaches precision 0.0102 at a crack prevalence of 0.26%. Recall is 0.6918 — the cracks are found — but far too much else is marked, and at this prevalence a small absolute false-positive area destroys precision. A method paper that shows only the sets where a method works is not reporting a method.

3.4 Against classical detectors, the detector does not lead

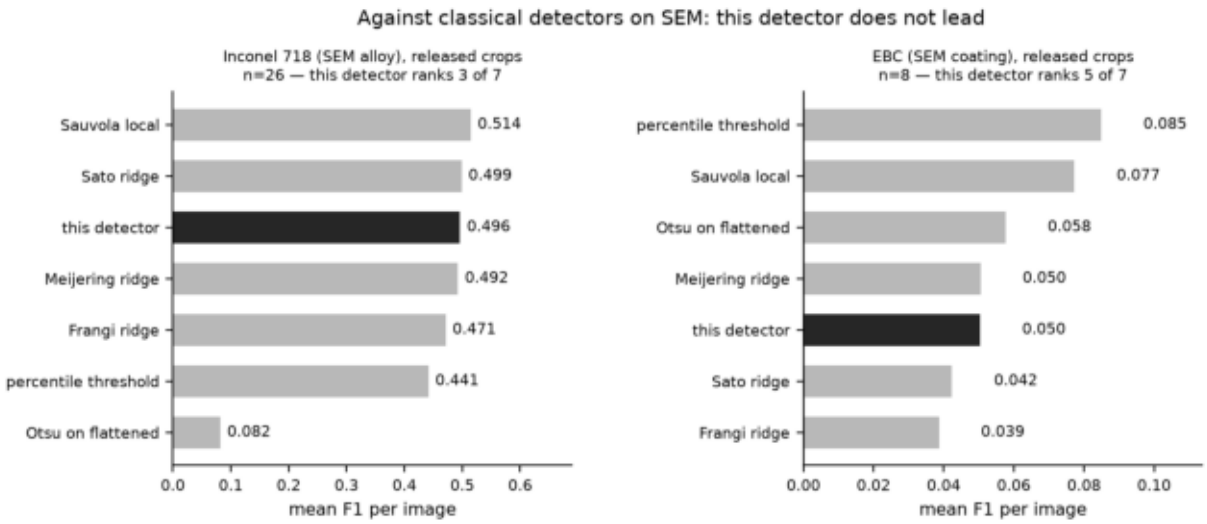


Figure 3. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same images, through the same scoring function. On the SEM alloy this detector ranks third of seven: Sauvola local thresholding reaches 0.5145 and Sato ridge 0.4985 against our 0.4958. On the coating it ranks fifth of seven.

The top three on the alloy are separated by 0.0186 F1, about the width of a gap this project has elsewhere called irresolvable at this sample size, so the honest statement is that on SEM alloy this detector and two classical filters are indistinguishable. We built it for micrographs and on micrographs it is matched by a local threshold from 2000 and a ridge filter from 1998.

This comparison was run on the released crops, the only set all seven arms have been scored on, so every arm in it is flattered by the same cropping described in §3.1.

4 What the scores measure

This is the part we would most want another group to check.

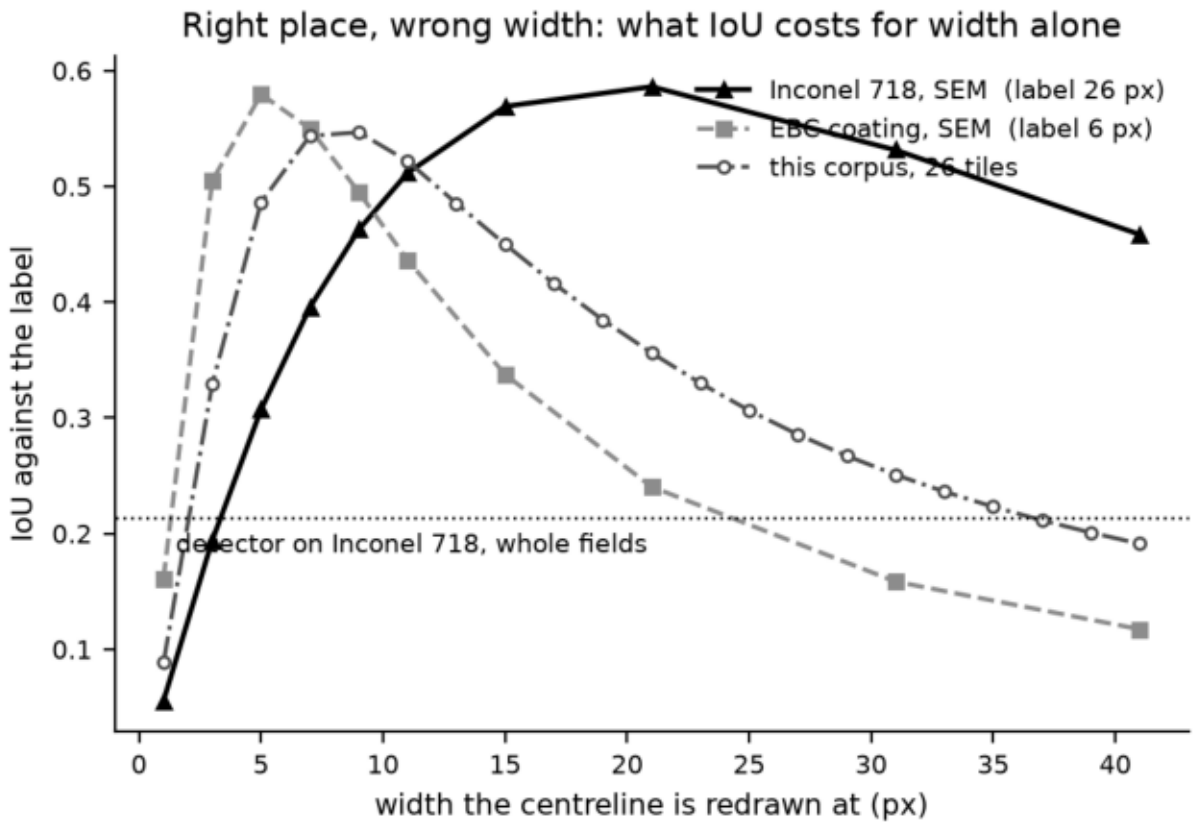


Figure 4. P3 - width sensitivity

Take the ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. Scored against the label it should be nearly perfect, and it is not. Most of the available IoU is being spent on agreeing about width, and the curves peak at different widths in different corpora — so the same detector's IoU can differ between datasets without its localisation differing at all.

The SEM sets make the point sharply on their own. Inconel 718's labels are drawn at a median 25.8 px and the coating's at 6.4 px — a factor of four apart — and the best constant-width reconstruction reaches 0.5861 and 0.5797 respectively. Four times the brush width, the same ceiling to within 0.007.

This does not set a ceiling, and we presented it as one. A real label varies in width along its length, so a detector that tracks that variation can score above a constant-width redraw of the same centreline. We had to be told this by the measurement.

And width does not explain the detector's difficulty here, which we initially claimed it did. The two SEM sets are annotated at a median 25.8 px and 6.4 px — four times apart — and their best constant-width reconstructions reach 0.5861 and 0.5797, the same ceiling to within 0.007. Our own laboratory corpus, scored by the same construction, peaks at 0.5467. Whatever makes these images hard for this detector, it is not how wide the labels are drawn.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants singly and jointly, plus every morphology and size floor, under leave-one-specimen-out

selection. No arm beat doing nothing on held-out data; the joint optimum lost 0.021 F1 when its selection was held out, and resolution-matching the scale ladder to the measured width ratio moved F1 by 0.0008.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of p = 0.125 — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

F1 is width-gameable where the labels are broad. On the same corpus a by-construction-correct 3 px crack scores F1 0.1260, below marking every pixel (0.1499), while F1 climbs to 0.64 on marked width alone. "Beats the no-information floor on F1" is necessary and not sufficient; precision lift separates where F1 does not.

Six precision rules validated on crops were deleters on frames. See §2.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the no-information floor beside every score. Marking every pixel scores F1 0.0351 on the SEM alloy and 0.0053 on the coating. A bare score cannot be read without it, and the floor moves by a factor of seven between those two sets alone.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Our laboratory corpus detects about +0.14 F1 for a new method; we reported smaller differences for weeks before measuring this.
- Controls must be able to fail. Every control we built was broken on purpose to check it noticed; two did not and were rewritten.
- Check the mask encoding and the bit depth before scoring anything. The two SEM crack sets use different mask conventions, and the images are 16-bit: an ordinary 8-bit conversion drove recall to 0.0012 and looked like a detector failure rather than a loading bug.
- Score whole fields, not crops of them. Cropping these images around their own cracks raises F1 by 0.1514 with no change to the method.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- The SEM evidence is 31 images of one alloy and 24 of one coating, from a single released collection whose record is one sentence long and names no pixel size, no microscope and no annotation protocol. Two crack classes out of eighteen targets. That is enough to place the detector and not enough to characterise it.
- What was searched for and not found. Public crack segmentation data in this modality is close to non-existent. Ruled out, each for a stated reason: road and pavement sets (CrackForest, DeepCrack, Crack500, CrackSeg9k and the aggregated collections) are the wrong modality; MetalDAM is SEM steel with pixel masks but its five classes are phases and a generic "defect", not cracks; NASA's MicroNet superalloy tasks label precipitates, not cracks; a released Ti-6Al-4V fractography set (SE and BSE, 16 GB) ships images with no masks at all; and the published nickel-superalloy fatigue-crack segmentation work we found is tomography with no data release. A second independent SEM crack set would change

this paper more than any further work on the method.

- Pavement results were measured and removed. The detector reaches F1 0.7932 and IoU 0.6809 on DeepCrack, ranking first of seven there. It is not reported above because a road-surface benchmark says nothing about micrographs, and the contrast between that ranking and the third-of-seven on SEM is better read as a warning about cross-modality benchmarking than as a result. The numbers remain in docs/figures/public_metrics.json.
- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.
- The corpus series in Figure 3 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.
- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Appendix A. The detector, end to end



Figure 5. Every constant read from analysis/detector_overrides.py when the plate was drawn.